Random Walk and Lévy Flight Theory App

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Abstract

A standard random walk models a particle that takes steps of fixed length in random directions. Over many steps, the displacement follows a Gaussian distribution and the mean-square displacement grows linearly in time — the hallmark of normal diffusion. Lévy flights generalise this picture by allowing step lengths to be drawn from a heavy-tailed distribution, meaning very large jumps occur far more often than a Gaussian would predict. The result is anomalous diffusion: the particle spreads through space much faster than a standard random walker, punctuated by occasional long-range relocations that dominate the overall trajectory. This behaviour appears across a broad range of natural and financial systems, from the foraging paths of animals to price movements in financial markets. The stability index μ ($0 < \mu \leq 2$) controls the tail heaviness — smaller values produce more extreme jumps and faster spreading, while $\mu = 2$ recovers the standard Gaussian random walk. This project presents an interactive browser-based tool, built with Streamlit, that lets users explore both processes side by side. Adjustable parameters feed into analytical PDFs, Monte Carlo simulations, scaling-collapse plots, and anomalous-diffusion exponent checks, with 2D animations providing an intuitive visual contrast between the compact cloud of a normal diffuser and the sparsely scattered, long-ranging pattern of a Lévy flier.

GitHub Repository

Source code and full project history are available at:

https://github.com/nandan9989/MSE5840-Project

How to Run

Prerequisites: Python 3.9+ with the following packages installed:

```
pip install streamlit numpy matplotlib scipy
```

Clone the repository and launch the app from the project directory:

```
git clone https://github.com/nandan9989/MSE5840-Project.git
cd MSE5840-Project
streamlit run app.py
```

The app will open automatically in your default browser at **http://localhost:8501**. Adjust parameters in the left sidebar and click **Run** to generate all plots and animations. SciPy is optional; if not installed, the Lévy Monte Carlo comparison section is skipped and all other features remain fully functional.

References

- Anomalous Transport: Foundations and Applications — Klages, R., Radons, G., & Sokolov, I. M. *Introduction to the Theory of Lévy Flights*, Chapter 5 in *Anomalous Transport: Foundations and Applications*. Wiley-VCH. https://doi.org/10.1002/9783527622979.ch5

- Lévy Flights in Random Environments — Fogedby, H. C. "Lévy Flights in Random Environments." *Physical Review Letters*, 73(19), 2517–2520, 1994. https://doi.org/10.1103/PhysRevLett.73.2517

- Claude (Sonnet 4.6) — Used for debugging assistance, website formatting, and visual feature refinement.